CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 25: Probability

Topic-wise short notes with solved examples

Probability is a number between 0 and 1 that measures how likely an event is. This chapter covers the probability of an event when all the outcomes are equally likely.

1 Basic terms

- A **random experiment** is one whose result cannot be predicted in advance, such as tossing a coin, throwing a die or drawing a card.
- Each possible result is an **outcome**. The set of all possible outcomes is the **sample space**, S.
- An **event** E is any set of outcomes. The outcomes in it are the **favourable outcomes**.
- Outcomes are **equally likely** when none is expected to occur more often than another.

2 Probability of an event

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} = \frac{n(E)}{n(S)}$$

- $0 \le P(E) \le 1$.
- An **impossible event** has probability 0; a **sure (certain) event** has probability 1.
- The event "not E" is the **complementary event** $\bar{E}$:

$$P(E) + P(\bar{E}) = 1$$

Example 2.1 A die is thrown once. Find the probability of getting (i) an even number (ii) a multiple of 3 (iii) a number less than 1 (iv) a number not greater than 6.

SOLUTION

$S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$

(i) $\{2, 4, 6\}$: $P = \frac{3}{6} = \frac{1}{2}$ (ii) $\{3, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$

(iii) No such number: $P = 0$ (iv) All six numbers: $P = 1$

(i) $\frac{1}{2}$ **(ii)** $\frac{1}{3}$ **(iii)** 0 **(iv)** 1

3 Coins and dice

Experiment	Number of outcomes	Sample space
One coin	2	H, T
Two coins	4	HH, HT, TH, TT
Three coins	8	HHH, HHT, HTH, THH, HTT, THT, TTH, TTT
One die	6	1, 2, 3, 4, 5, 6
Two dice	36	Ordered pairs $(1,1)$ to $(6,6)$

Example 3.1 Three coins are tossed together. Find the probability of getting (i) exactly two heads (ii) at least one tail.

SOLUTION

$n(S) = 8$

(i) HHT, HTH, THH: $P = \frac{3}{8}$

(ii) "At least one tail" is the complement of "no tail" (HHH): $P = 1 - \frac{1}{8}$

(i) $\frac{3}{8}$ (ii) $\frac{7}{8}$

Example 3.2 Two dice are thrown together. Find the probability that (i) the sum is 7 (ii) the two numbers are equal (iii) the sum is more than 10.

SOLUTION

$n(S) = 36$

(i) $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$: $P = \frac{6}{36} = \frac{1}{6}$

(ii) $(1,1), (2,2), \ldots, (6,6)$: $P = \frac{6}{36} = \frac{1}{6}$

(iii) $(5,6), (6,5), (6,6)$: $P = \frac{3}{36} = \frac{1}{12}$

(i) $\frac{1}{6}$ (ii) $\frac{1}{6}$ (iii) $\frac{1}{12}$

4 Playing cards

- A pack has **52 cards**: 4 suits of 13 cards each.
- Spades and clubs are **black** (26 cards); hearts and diamonds are **red** (26 cards).
- Each suit has an ace, the numbers 2 to 10, a jack, a queen and a king.
- **Face cards**: jack, queen and king. There are 12 face cards, 6 red and 6 black.

Example 4.1 A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is (i) an ace (ii) a black king (iii) a face card (iv) neither a heart nor a king.

SOLUTION

(i) 4 aces: $P = \frac{4}{52} = \frac{1}{13}$ (ii) 2 black kings: $P = \frac{2}{52} = \frac{1}{26}$

(iii) 12 face cards: $P = \frac{12}{52} = \frac{3}{13}$

(iv) Hearts or kings: 13 hearts + 3 other kings = 16 cards, so $52 - 16 = 36$ cards: $P = \frac{36}{52} = \frac{9}{13}$

(i) $\frac{1}{13}$ (ii) $\frac{1}{26}$ (iii) $\frac{3}{13}$ (iv) $\frac{9}{13}$

5 Balls, numbered cards and letters

Example 5.1 A bag contains 4 red, 6 blue and 5 green balls. One ball is drawn at random. Find the probability that it is (i) blue (ii) not green (iii) red or green.

SOLUTION

Total = 15 balls

(i) $P = \frac{6}{15} = \frac{2}{5}$ (ii) $P = \frac{10}{15} = \frac{2}{3}$ (iii) $P = \frac{9}{15} = \frac{3}{5}$

(i) $\frac{2}{5}$ (ii) $\frac{2}{3}$ (iii) $\frac{3}{5}$

Example 5.2 Cards numbered 1 to 25 are placed in a box and one is drawn at random. Find the probability that the number is (i) prime (ii) a perfect square (iii) divisible by both 2 and 3.

SOLUTION

(i) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, which is 9 numbers: $P = \frac{9}{25}$

(ii) 1, 4, 9, 16, 25: $P = \frac{5}{25} = \frac{1}{5}$

(iii) Multiples of 6: 6, 12, 18, 24: $P = \frac{4}{25}$

(i) $\frac{9}{25}$ (ii) $\frac{1}{5}$ (iii) $\frac{4}{25}$

Example 5.3 A letter is chosen at random from the word PROBABILITY. Find the probability that it is (i) a vowel (ii) the letter B.

SOLUTION

The word has 11 letters: P, R, O, B, A, B, I, L, I, T, Y.

(i) Vowels: O, A, I, I, which is 4 letters: $P = \frac{4}{11}$

(ii) B appears twice: $P = \frac{2}{11}$

(i) $\frac{4}{11}$ (ii) $\frac{2}{11}$

6 Finding an unknown number

Example 6.1 A bag contains 12 balls, of which x are white. (i) Write the probability of drawing a white ball. (ii) If 6 more white balls are added, the probability of drawing a white ball doubles. Find x.

SOLUTION

(i) $P = \frac{x}{12}$

(ii) Now there are $x + 6$ white balls out of 18: $\frac{x+6}{18} = 2 \times \frac{x}{12}$

$\frac{x+6}{18} = \frac{x}{6}$, so $x + 6 = 3x$

$x = 3$

7 Common mistakes

- Writing a probability greater than 1.
- Treating $(3, 4)$ and $(4, 3)$ as the same outcome for two dice.
- Counting 1 as a prime number.
- Counting the king of hearts twice in "a heart or a king".
- Leaving the answer unsimplified.

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